Rodrigo Aguiar

Senior Engineer | AI & Data Engineer

Jaraguá do Sul, SC, Brazil (GMT-3) — Open to remote opportunities
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PROFESSIONAL SUMMARY

Senior Engineer with 20+ years of experience bridging Industrial R&D, Data Science, and Systems Architecture. Currently focused on building AI Agents, orchestrating LLM pipelines (RAG), and implementing MLOps workflows. Proven track record of delivering 80%+ efficiency gains and generating measurable cost savings through data-driven engineering solutions. Open to remote opportunities with full availability for US time zone alignment.

CORE SKILLS

Languages: Python (Advanced), SQL

AI/LLM Stack: LangChain, RAG, Ollama, Google Gemini API, OpenAI API

Data Science: Pandas, NumPy, Scikit-Learn, Statsmodels, Predictive Modeling

Engineering & DevOps: FastAPI, Docker, OCI, CI/CD, Git

Visualization: Power BI, Matplotlib, Seaborn

Standards & Compliance: ISO 17025, ISO 9223/12944, ASTM Standards

Quality Management: Six Sigma Black Belt

PROFESSIONAL EXPERIENCE

Technology Advisor — Independent

2026 – Present

Advise companies on the implementation of digital tools, process automation, and data analysis. Creator of the RSA Method (Reorganize, Simplify, and Automate), helping businesses eliminate repetitive manual tasks through modern technological solutions and AI Agents.

R&D Researcher — Multinational Corporation

2015 – 2025

- Led the end-to-end implementation of a LIMS and materials management platform (Ansys Granta MI & Selector), increasing technical data reliability under ISO/IEC 17025 and reducing internal client response time by over 80%.
- Developed polymer reliability testing protocols using thermal analysis (ASTM/ISO standards), generating savings of over $40,000 per material while cutting result delivery time to 3 days.
- Authored the Technical Paint Equivalence Manual (ISO 9223 and 12944), unifying technical criteria across Quality, Technical Support, and Engineering teams.
- Built Python-based automation pipelines (Pandas, NumPy, Statsmodels, Scikit-Learn) to eliminate repetitive tasks and model industrial process data.

EDUCATION

MSc, Chemical Engineering — UFSC

2013 – 2015

Specialized in Second-Generation Bioethanol (E2G). In silico investigations using Metabolic Control Analysis (MCA).

BSc, Chemical Engineering — UFPR 2006 – 2012

Solid foundation in industrial processes, thermodynamics, and chemical kinetics.

PUBLICATIONS

Kinetics of enzyme-catalyzed hydrolysis of steam-exploded sugarcane bagasse (ScienceDirect)

Assessment of the enzymatic hydrolysis profile of cellulosic substrates based on reducing sugar release (ScienceDirect)

CERTIFICATIONS

Data Science (Credly) · Neural Networks and Deep Learning (Coursera) · Six Sigma Black Belt

PROJECTS IN PROGRESS

Central Market Agent — Autonomous RAG agent for corporate knowledge retrieval (LangChain, Ollama)

GenAI Insurance Bot — AI-powered customer service assistant (FastAPI, ChromaDB, Gemini)